
Story2Image: Story-Grounded Latent Diffusion for Coherent Visual Narratives

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1 Introduction

Generating coherent visual-textual narratives is a formidable challenge that requires maintaining fine-grained semantic consistency across disparate modalities. This task involves mapping a brief input prompt x to a dual-output $(y_{\text{story}}, I_{\text{image}})$, where y_{story} is a long-form narrative and I_{image} is a representative illustration. The fundamental difficulty lies in ensuring that I_{image} captures the specific narrative nuances introduced during the expansion of x into y_{story} .

To address this, I present **Story2Image**, a cascaded, two-stage framework designed to transform short textual prompts into detailed narratives accompanied by visually aligned imagery. This model employs a pre-trained autoregressive language model for story expansion (1), followed by a latent diffusion model (2) that synthesizes a coherent image conditioned on the story’s semantic embeddings via cross-attention. In this report, I describe the design of this architecture, formalize the training objectives for both stages, and outline a practical alignment strategy to maximize cross-modal consistency through fine-tuning.

2 Proposed Architecture

The Story2Image framework consists of two sequential modules: an Autoregressive Story Expander and a Story-Conditioned Latent Diffusion Model.

2.1 Autoregressive Story Expansion

I utilize a pre-trained Transformer-based decoder (1) to model the conditional probability of a narrative $y_{\text{story}} = (y_1, y_2, \dots, y_T)$, where T denotes the total number of tokens. Given a short prompt x , the model predicts the sequence token-by-token. Let θ represent the model parameters; the training objective is to minimize the standard negative log-likelihood (cross-entropy loss):

$$\mathcal{L}_{\text{text}}(\theta) = - \sum_{t=1}^T \log P(y_t \mid y_{<t}, x; \theta) \quad (1)$$

This formulation ensures that the generated story remains linguistically coherent while logically expanding upon the initial premise provided by x .

2.2 Story-Conditioned Latent Diffusion

For visual synthesis, I adopt a Latent Diffusion Model (LDM) (2), which operates in the compressed latent space of a pre-trained Autoencoder. The story y_{story} is first mapped to a continuous semantic embedding $c = \text{Encoder}(y_{\text{story}})$ using a frozen CLIP text encoder (3).

The diffusion process involves a forward pass that adds Gaussian noise ϵ to a latent representation z_0 over K timesteps, and a reverse denoising pass. I implement the conditioning mechanism via cross-attention layers within a U-Net noise predictor ϵ_ϕ . The model learns to predict the noise injected at timestep k by minimizing the denoising score matching objective:

$$\mathcal{L}_{\text{diff}}(\phi) = \mathbb{E}_{z \sim \mathcal{E}(I), \epsilon \sim \mathcal{N}(0, \mathbf{I}), k, c} [\|\epsilon - \epsilon_\phi(z_k, k, c)\|_2^2] \quad (2)$$

where $\mathcal{E}$ is the VAE encoder, z_k is the noisy latent at step k , and ϕ represents the U-Net parameters. By injecting c into the cross-attention layers, the visual generation process attends to specific narrative elements in y_{story} .

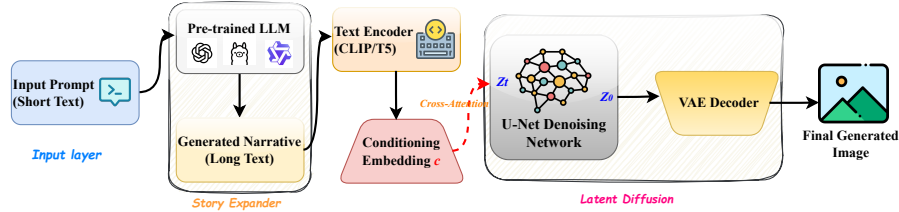


Figure 1: Overview of the Story2Image architecture. The framework pipelines a short prompt through an LLM to generate a narrative, which then conditions a Latent Diffusion Model via cross-attention to produce a consistent image.

3 Training Strategy

To achieve effective alignment, I propose a two-stage training strategy. In the first stage, the Story Expander and the LDM are trained independently on large-scale text and image-caption datasets to learn foundational generation capabilities. In the second stage, I perform joint alignment tuning. I freeze the Story Expander and the text encoder to preserve their semantic prior. I then fine-tune only the cross-attention layers of the LDM U-Net using a curated dataset of story-image triples containing the short prompt x , the generated story y_{story} , and the ground-truth image I_{GT} .

4 Example Generation

I demonstrate the framework’s capability by transforming a fantasy-themed prompt into a detailed story and a corresponding image.

Input Prompt (Short Description)

"A lonely wizard discovers a dragon egg in a volcano."

Output 1: Generated Story (Long-form Text)

"Deep within the obsidian heart of Mt. Ignis, Silari, a witch exiled for her forbidden pyromantic studies, wandered through rivers of molten glass. Near the primary caldera, she stumbled upon an anomaly: a heavily scaled egg, dark as midnight yet pulsing with a rhythmic, crimson heat. As Silari reached out a trembling hand, the shell fractured with a crystalline crack, and a small, ruby-red dragon hatchling tumbled into the ash. Its eyes, glowing like twin embers, fixed upon the witch, igniting a primal magic that promised to change the fate of the realm forever."

Output 2: Generated Image



Table 1: Example generation showing semantic alignment between the long-form narrative and visual details.

References

- [1] Vaswani, A., et al. (2017). Attention Is All You Need. *NeurIPS*.
- [2] Rombach, R., et al. (2022). High-Resolution Image Synthesis with Latent Diffusion Models. *CVPR*.
- [3] Radford, A., et al. (2021). Learning Transferable Visual Models from Natural Language Supervision. *ICML*.